

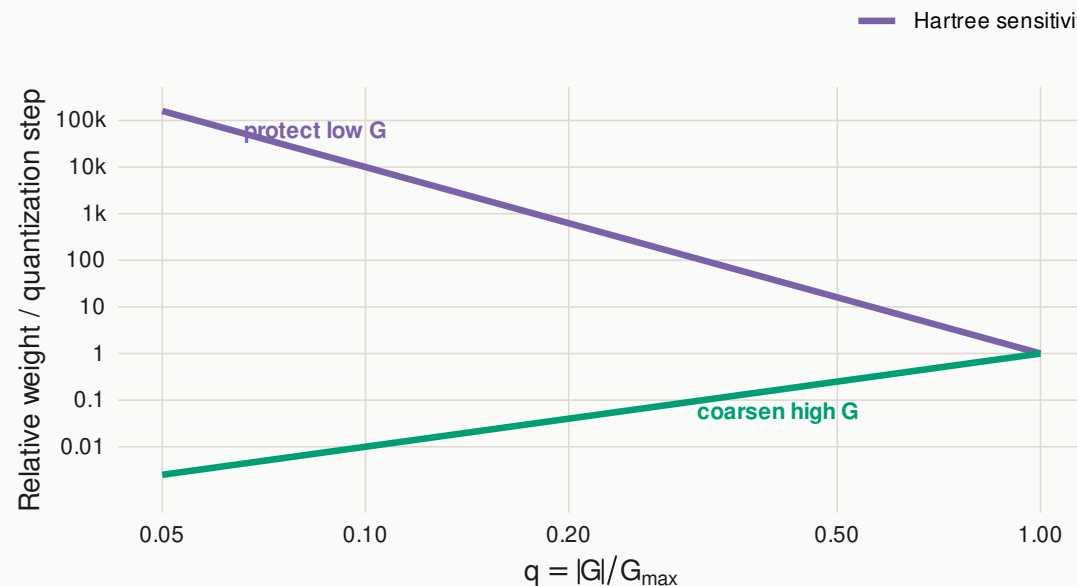
Figure 8 | The downstream operator converts compression diagnosis into design

QOAC-H allocates reciprocal-space error from the Hartree operator and verifies the allocation against an operator-blind ablation.

The frozen design is then tested on a disjoint cohort and the complete development population.

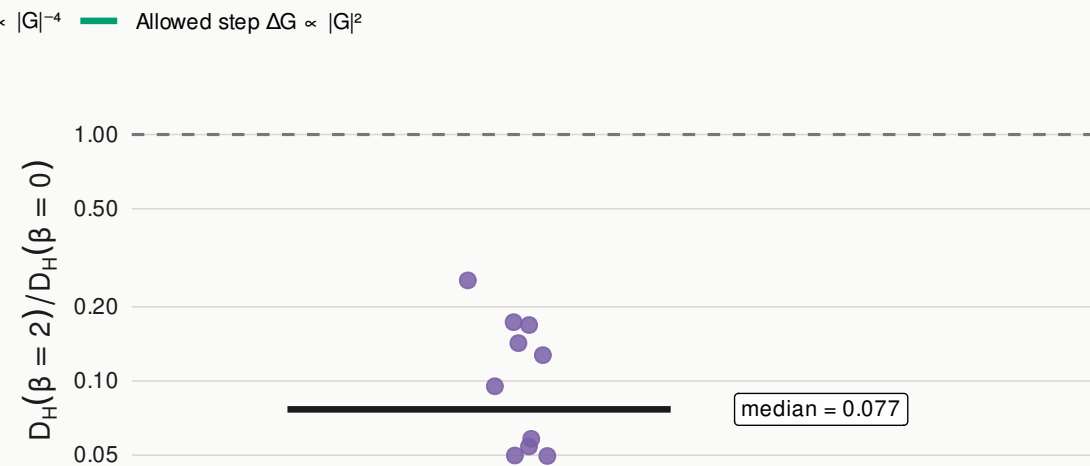
A | The Hartree operator defines the distortion geometry

Normalized at $q = 1$; the quantizer is derived from the physical Fourier weighting



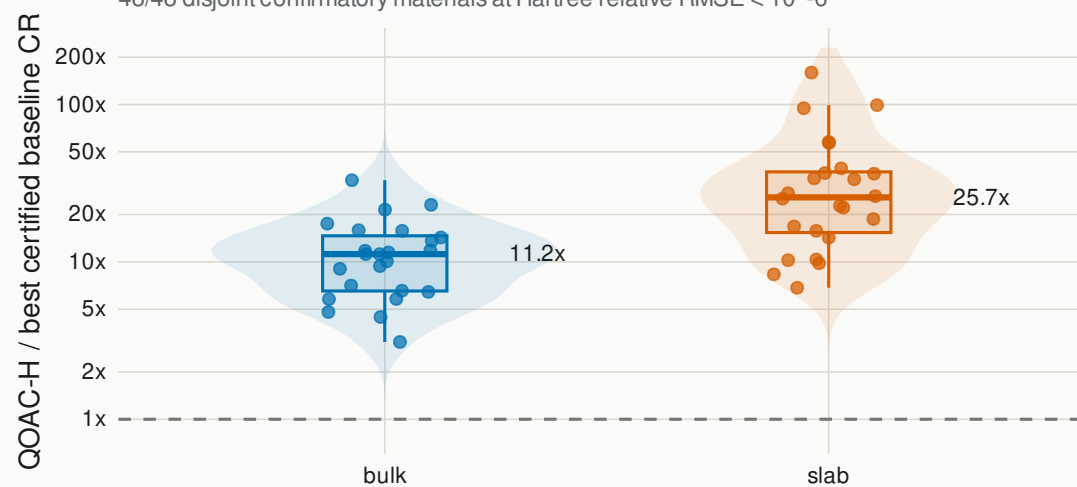
B | Operator-derived $\beta=2$ beats operator-blind $\beta=0$

12/12 engineering materials; matched serialized storage



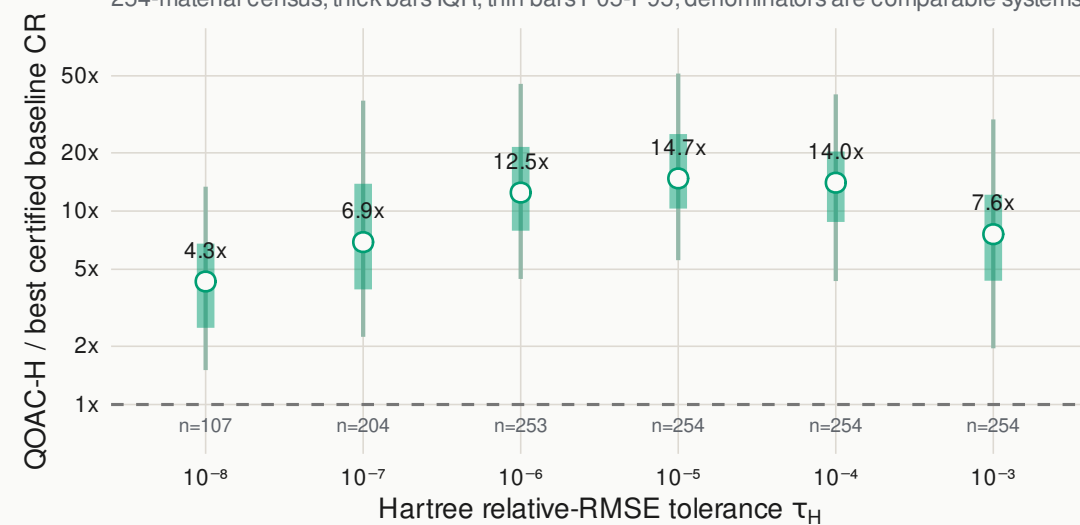
C | The frozen codec wins on unseen materials

48/48 disjoint confirmatory materials at Hartree relative RMSE $< 10^{-6}$



D | The gain persists across the full Hartree contract ladder

254-material census; thick bars IQR, thin bars P05-P95; denominators are comparable systems



Primary Hartree contract: relative RMSE $< 10^{-6}$. Confirmatory cohort: 48/48 wins; median CR ratio 15.016x (bootstrap 95% CI 11.204–21.461); 48/48 Nyquist-safe guardrails passed.

Full census at 10^{-6} : 253/253 comparable wins; median 12.463x; P05 4.459x; minimum 2.444x. Eligibility and baseline availability define the denominator at each tolerance.